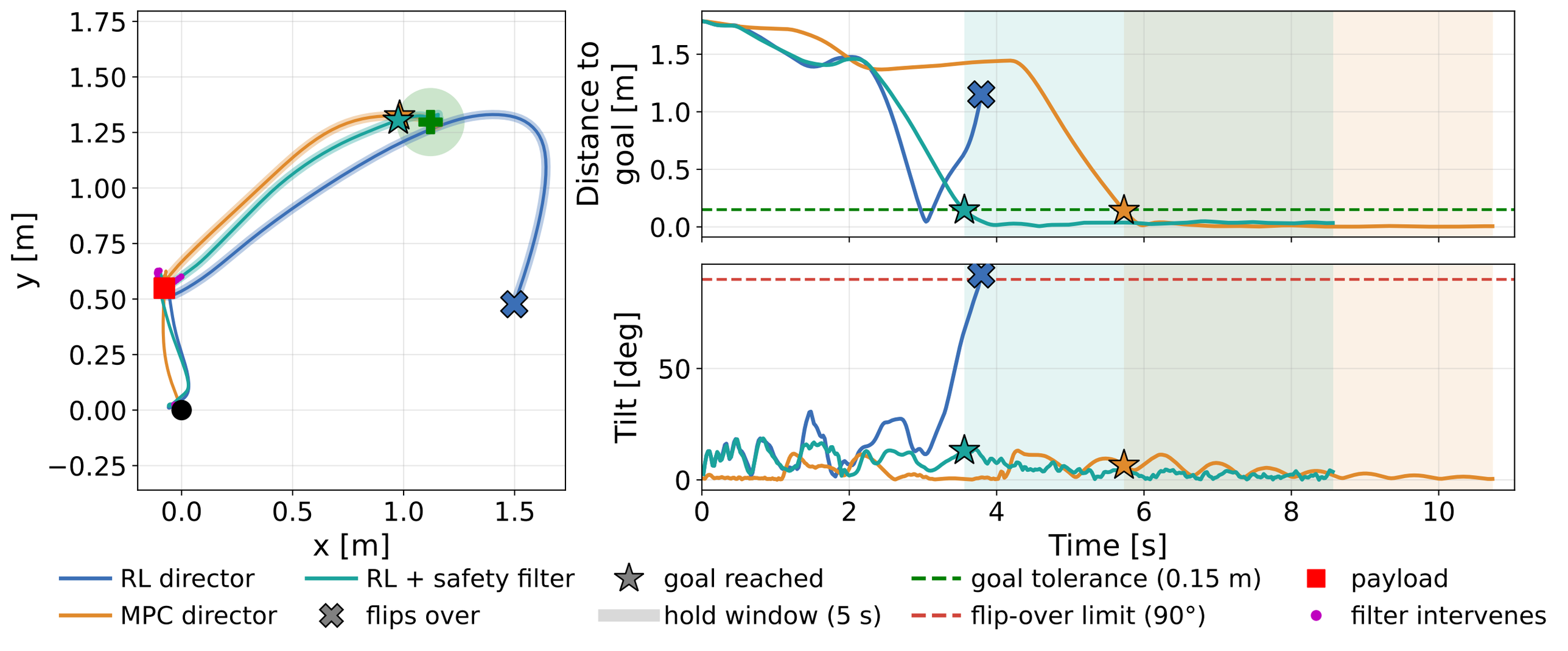
y [m]
x [m]
Distance to goal [m]
Tilt [deg]
Time [s]
RL director
MPC director
RL + safety filter
flips over
goal reached
hold window (5 s)
goal tolerance (0.15 m)
flip-over limit (90°)
payload
filter intervenes